

POWERFACTORY

PowerFactory 2024

Technical Reference

WECC Battery Energy Storage System Templates

WECC BESS 1.0MVA 50Hz,
WECC BESS 1.0 MVA 60Hz

PF2024

POWER SYSTEM SOLUTIONS
MADE IN GERMANY

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Contents

1 Introduction **1**

2 Model Usage **2**

3 Model Configuration **2**

3.1 Basic Configuration 2

3.1.1 Adjusting Rated Power 2

3.1.2 Adjusting Power Dispatch 2

3.1.3 Model Aggregation 2

3.2 Dynamic Model Configuration 3

4 Model Documentation **3**

4.1 BESS 3

4.2 Measurement 4

5 Plant Controller **5**

6 References **6**

List of Tables **7**

1 Introduction

This document describes the usage of the WECC BESS (battery energy storage system) models in *PowerFactory*. The models are based on [3, 8].

There are two WECC BESS templates available in PowerFactory which is configured for 50 Hz and 60 Hz systems. Each template has a rated apparent power of 1 MVA.

Chapters 2 and 3 describe how to use and configure the models in *PowerFactory*. In chapter 4 the different models are described.

Note: The models which are available in *PowerFactory* as templates are configured with a set of sample parameters. The user has to adjust these parameters (i.e. power rating, voltage level, reactive power limits and all parameters of the dynamic models) to represent a specific BESS of a specific manufacturer!

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2 Model Usage

The template consists of a static generator (*ElmGenstat*), a terminal, a two winding transformer and of a composite model which defines the dynamics of the BESS. The composite model contains the dynamic models using DSL (*DIgSILENT* Simulation Language) and all necessary measurement devices.

For each template generic protection devices are provided to act in case of over or under frequency and voltages during operation. The templates are accessible to be used in a project through the button *General Templates* in the drawing toolbox.

3 Model Configuration

The rated power, dispatched power and number of parallel converters can be adjusted in the templates considering the next changes.

3.1 Basic Configuration

The following sub-chapters describe the basic configuration, needed for all kind of calculations.

3.1.1 Adjusting Rated Power

The rated power of the static generator has to be configured on the *Basic Data* page of the element itself (i.e. the static generator doesn't need a type). The rated active power is calculated via the nominal apparent power "sgn" and the power factor "cosn".

3.1.2 Adjusting Power Dispatch

The dispatch of the static generator can be entered on the *Basic Data* page. The default bus type is "PQ". Using this bus type allows to enter the active and reactive power dispatch. The reactive power can be limited by the reactive power capability curve which can be also voltage dependent. It is also possible to coordinate the reactive power output of several static generators by using a station controller (*ElmStactrl*). Positive power values represent a discharging battery, negative values on the other hand a charging battery.

3.1.3 Model Aggregation

The WTG can also be used for aggregated representation of several WTGs or even a whole wind farm. The only setting needed for this is the "Number of parallel machines" on the *Basic Data* page of the element. The active power output of the generator is then the active power dispatch times the number of parallel machines. The set-up transformer has to be adjust accordingly.

3.2 Dynamic Model Configuration

The dynamic simulation is started from steady state, which is the load flow result. It is therefore important to check the load flow configuration of the elements.

The dynamic model can either be accessed via the generator element in the single line diagram or via the data manager.

- Via Element: Open the element. Go to the *Basic Data* page and click on the arrow button after “Model”.
- Via Data Manager: Search the corresponding composite model , right click and select “Edit”.

After opening the composite model the different dynamic (common) models can be accessed via a double click in the 2nd column.

Note: The models which are available in *PowerFactory* as templates are configured with a set of sample parameters. It is task of the user to adjust these parameters (i.e. power rating, voltage level, reactive power limits and all parameters of the dynamic models) to represent a specific BESS of a specific manufacturer!

The dynamic (common) models can be configured via scalar input parameters. These parameters are displayed after opening a common model on the *General* page.

The “REEC_C Electrical Control Model” contains also arrays (i.e. a list of parameters). The arrays can be found on the second page of the common model *Advanced 1*. Each array consists of two columns. The dimension can be changed by modifying the number in the top left cell of an array. The minimum dimension is two. The page has to be refreshed after changing the dimension by switching once to the “General” page and then back to “Advanced 1”.

4 Model Documentation

In the following sub-sections the different types are described. In Table 4.1 the different dynamic models are listed.

DSL Model	Block Definition	BESS
REEC_C Electrical Control Model	REEC_C	X
REGC_B Generator-Converter Model	REGC_B	X
WTGWGO_A Weak Grid Option Model	WTGWGO_A	X
Protection Model	Protection	X
Plant Control	Frame WECC Plant Control	(X)

Table 4.1: Block Overview

4.1 BESS

The BESS is represented in the network via the static generator element. The initial settings for the load flow calculation and dynamic simulation are done in the static generator, configuration

of the grid element is described in section 3.

The generic BESS as build in *PowerFactory* consist of the following components:

- Generator.
- REEC_C, which implements the electrical control model. The main task is the control of the reactive power. Different control models are available as shown in Table 4.2. It supports also a minimal battery representation in form of an integrator.
- REGC_B, which implements the fast converter controllers and represents the interface between the generator/converter [8].
- WTGWGO_A, which implements a supplemental weak grid option control [8].
- Protection, which implements the protection functions with disconnection from the grid in case of over- or under- frequency, over- or under- voltage.
- Plant Control. This implements the plant controller which calculates the commanded active and reactive current. The plant control is available as template in the templates folder "Plant Controllers", see also chapter 5.
- Voltage and power measurement.

Control Mode	**PfFlag**	**Vflag**	**Qflag**
Local constant Q control	0	1	0
Local constant pf control	1	1	0
Local voltage control	0	0	1
Local coordinated Q/V control	0	1	1

Table 4.2: Electrical control modes [2]

Note: The REGC_B model is by default applied in the BESS templates; however, the REGC_A model can be used instead if desired [5]. The REGC_A model type is available within the DIgSILENT Library/Dynamic Models.
The REGC_B model is defined as a controlled voltage source [8]. The former REGC_A model was defined as a controlled current source [1]. The static generator element is configured to be a model *according to connected input signals* in RMS simulation, i.e. to represent a controlled voltage source or a controlled current source depending on the REGC model used.
If the REGC_A model is used, *DIgSILENT*recommends to use the additional built-in option *dq-reference angle delay* of the static generator for better numerical robustness, especially when simulating deep voltage dips. This option is described in detail in the technical reference paper of the static generator element [4]. The *delay time constant* should correspond to a typical time constant of a PLL.

4.2 Measurement

Power and voltage measured signals are implemented in the models using the *PowerFactory* inbuilt elements *StaPqmea* and *StaVmea* correspondingly [6], [7].

In all models the frequency signal needed in the Protection Model is taken from the voltage measurement device *StaVmea*, please read the technical reference paper of this device to learn more about the frequency calculation.

5 Plant Controller

A plant level controller template as specified in [3] is also provided. The “WECC Plant Controller” template is accessible using the button *General Templates* in the drawing toolbox.

The plant controller allows to control either voltage, with droop or line drop compensation, or reactive power. Additionally an active power control function including a simple droop-down function in case of over-frequencies is also included, as optional feature an up regulation function is incorporated too. A dedicated flag can be set to enable or disable active power control.

This model is intended to be used with aggregated plant representation only.

6 References

[1] WECC Second Generation Wind Turbine Models, January 2014.

[2] Model User Guide for Generic Renewable Energy System Models. Technical Report 3002006525, EPRI, June 2015.

[3] WECC Energy Storage System Model - Phase II, April 2015.

[4] *Technical Reference "Static Generator"*. DIgSILENT GmbH, Gomaringen, Germany, 2019.

[5] WECC Brief Summary of all 2nd Generation Generic Renewable Energy System Dynamic Models, February 2021.

[6] *Technical Reference "Power Measurement Device"*. DIgSILENT GmbH, Gomaringen, Germany, 2023.

[7] *Technical Reference "Voltage Measurement Device"*. DIgSILENT GmbH, Gomaringen, Germany, 2023.

[8] P. Pourbeik. Proposal for new features for the renewable energy system generic models 07/23/18 (revised 8/3/18; 11/14/18; 11/18/18; 12/13/18; 1/20/19; 2/7/19; 3/5/19; 6/24/19; 6/28/19; 7/3/19; 7/17/19; 8/19/19; 12/17/19; 8/14/20; 8/24/20;8/31/20; 11/12/20; 4/6/21; 4/15/21; 9/22/21; 11/11/21; 9/19/22). 2022.

List of Tables

4.1 Block Overview 3

4.2 Electrical control modes [2] 4